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Robust Vehicle Speed Estimation Based on Vision Sensor Using YOLOv5 and DeepSORT

Dea Angelia Kamil, Wahyono, and Agus Harjoko^(✉)

Department of Computer Science and Electronics, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia

dea.angelia.kamil@mail.ugm.ac.id, {wahyo,aharjoko}@ugm.ac.id

Abstract. For driving safety purposes, the policymakers make regulations on the speed limit on the highway because vehicles high speed may cause an accident. Vision-based vehicle speed estimation requires limited human resources and minimizes human error. Therefore, the authors of this research propose a vehicle speed estimation pipeline using You Only Look Once version 5 (YOLOv5) for vehicle detection, tracking vehicles using Simple Online and Realtime Tracking with a Deep Association Metric (DeepSORT), and speed measurement process. The dataset built for this work is taken in Gadjah Mada University using a camera in bird view orientation. The detection precision and recall of the of YOLOv5 pre-trained model used are both about 100% and has fps 10.762. The speed value reaches a Root Mean Square Error (RMSE) of 3.926 and a Mean Absolute Error (MAE) of 3.155. The RMSE obtained is compared with previous research and has the most significant value.

Keywords: Vehicle speed estimation · YOLOv5 · DeepSORT · Computer vision · Intelligent transportation system

1 Introduction

Driving safety scheme is improved with the popularity and development of the autonomous vehicle and part of Intelligent Transportation Systems: traffic density classification [1], vanishing point detection [2], illegally parked vehicles detection [3], traffic sign recognition [4], and vehicle classification [5]. One part of this scheme is vehicle speed detection. Not only do autonomous vehicles have to control the level of speed, but conventional vehicles also need to be managed. It is because vehicles moving at high speed may cause an accident. Therefore, the law enforcer makes the policy about managing the speed limit. This regulation may be applied on the highway. In fact, to this regulation, the speed detector is usually installed on the side road to detect the vehicle over the speed limit. A speed gun or radar gun is a frequently used tool for detecting vehicle speed. This method is based on non-intrusive technology [6]. There is much research using speed guns for obtaining ground truth information or to be the primary approach for detecting speed itself [7–10]. However, speed gun has disadvantages, and it depends on human power, which may cause human error. Radar guns have a simple

method to implement. People need to shoot a speed gun at the vehicle they want to detect the speed. However, human hands and eyes speed cannot handle it. The other technology is the intrusive-based method using an inductive loop detector. Luvizon et al. use this tool to generate the ground truth of their dataset [6]. This method avoids human error but has complicated installation. An inductive loop detector is installed under the asphalt and is sensitive to water. Moreover, the two leading technologies above are expensive.

Because of the problem mentioned, the authors proposed vehicle speed estimation based on computer vision technology. The joint vision-based speed estimation is divided into three main processes. Step one is detecting and classifying vehicles. The vehicle shown in the video is detected using a detector algorithm. The learning-based method is commonly used for detecting vehicles in speed estimation schemes such as SSD [11], YOLO family; YOLOv3 [12], YOLOv4 [12], Faster RCNN [13, 14], and Mask RCNN [15]. In this study, YOLOv5 is utilized for detecting vehicles. Step two is vehicle tracking. One of the famous tracking algorithms is Kalman Filter. Single Online and Realtime Tracking (SORT) [15, 16] is a tracker based on Kalman Filter and the Hungarian method for predicting motion. In this research, DeepSORT is used [15, 17]. After localization of the car, tracking is needed to track the bounding boxes in the whole appearance of the video. Furthermore, the distance and time travel ratio of vehicles passing the RoI is used for computing the vehicle velocity.

The following sections will cover the whole process of this research. Section 2 discusses the previous research on vehicle speed estimation using three-based processes, including detection, tracking, speed measurement, and learning-based approaches. Section 3 describes the proposed method's experimental, starting with how the dataset was collected, the experimental on object detection by YOLOv5, tracking using DeepSORT, and the method implemented for measuring vehicle speed. Section 4 presents the conclusion of the result, the weakness of the research, and the scheme of the future study.

2 Related Works

2.1 Learning-Based

The learning-based approach is rarely used for detecting vehicle speed because of the lack of data that has vehicle speed information in the video dataset. Generally, the learning-based method [18–21] discusses how to map the sequence of the image into regression information, in this case, speed. To realize the concept mentioned, Martinez et al. [20] adapted two different deep learning models: 3D CNN and CNN-RNN. The author generated dataset from synthetics driving simulator. Adapted two deep learning models: 3D CNN and CNN-RNN. The author generated a dataset from a synthetic driving simulator. However, due to the small number of data, the model needs to be more robustly explored whether environmental conditions influence the model. In the following research [21], the author increased the amount of data, and the experiment result indicates that if the model the author used, 3D CNN, was trained in a large dataset, it could be adequate to detect the speed of traveling vehicles with more outstanding results.

2.2 Three-Based Process

At least the process of getting vehicle speed value consists of the vehicle detecting, multi-object tracking, and estimating speed. In the part of vehicle detection influenced by the popularity of deep learning, the amount of research implementing deep learning for detecting and classifying vehicles is shown in the video data for the first step of the vehicle speed estimation scheme [12–17, 22]. The work of contribution to this step can be seen in the research of Lin et al. [12]. They compared YOLOv3 and YOLOv4 to detect and classify vehicles with an accuracy of about 98.02% and 99.50%, respectively. Furthermore, in the pipeline of estimating speed Inter-frame difference method is applied [20, 21]. This method consists of binary logical information and morphological information. Not only utilizing the inter-frame difference method but Trivedi et al. [23] also employ blob analysis to detect objects like in [6, 14]. However, in blob analysis, the different colors of the vehicle need to be sufficiently recognized. Still, in vehicle detection, Luvizon et al. [6] extracted features from vehicle movement using MHI (Motion Histogram Image) and matched the motion feature by SIFT (Scale Invariant Feature Transform).

The following process of detecting vehicle speed is tracking. The amount of research employed Kalman Filter for the base of the tracking object [14–17]. Sochor et al. [14] applied Kalman Filter to track a feature group of blobs representing one car. Moreover, SORT [15, 16] and DeepSORT [15, 17] are the famous trackers used in vehicle speed measurement studies. SORT and DeepSORT employ Kalman Filter to predict the bounding box for comparison with the detected one. If the Kalman Filter is used for predicting the localization of an object, Kanade-Lucas, and Tomasi usually use the corner feature. In [6], after the extracted feature, the author applies Kanade Lucas Tomasi to track the feature to produce the motion vector. The last process is measuring vehicle speed. In general, we divide the technique of measuring speed into calibrated [6, 14–16] or uncalibrated [7, 12, 13, 17, 24]. Detailed information on vehicle speed measurement will be described in Sect. 2.3.

2.3 Speed Estimation

Mostly, the vehicle speed estimation technique without calibrating the camera inputs the distance information obtained from the laser meter or meter wheel in the between of line or region of interest. Then, the distance information operates with the pixel information. Hence, the velocity values get from the distance and time travel ratio. In [13], the author uses the warping technique, so the image seen in the eagle view and the road is straight and flat. This technique makes the measurement process more robust and does not need camera distortion. On the other hand, the calibration-based technique uses camera parameters to obtain image orientation for accurate distances. In the research by Kumar et al. [15], the authors used an affine rectification technique to calibrate the camera. This method uses using Homography matrix to get intrinsic and extrinsic camera parameters. After calculating the homography operation, the distance information can be computed. In [25], the authors used the frame rate information and starting and endpoint of the region of interest (RoI), in this research they utilized white dashed detection to localize the RoI using color thresholding, for estimating vehicle speed.

3 Proposed Method

The method proposed in this study is to estimate vehicle speed using a traffic video dataset. The three-based process is adopted to detect the vehicle speed: vehicle detection, vehicle tracking, and speed estimation. The whole process is included in Fig. 1.

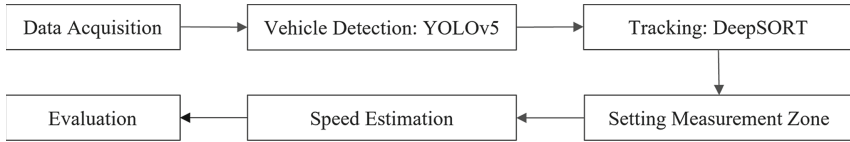


Fig. 1. Speed estimation process

3.1 Data Acquisition

The author collects the data built for this research from the veterinary faculty overpass of Gadjah Mada University, Indonesia. The video was captured using Canon 1200D. The testing video has 12 min long with 30 fps and the resolution is 1920×1088 . The field of view of the camera setting is bird view. A frame of the video dataset is illustrated in Fig. 2.



Fig. 2. Image of traffic video dataset (left) Bushnell Speed Gun (right)

For the evaluation process, the ground truth value of vehicle speed is obtained by speed gun, Bushnell Speed Gun/Radar Gun Velocity, from above the overpass. The image of the radar gun is shown in Fig. 3. Because of the limitation of human power to catch the vehicle speed, we just captured one class of vehicle that is the car for this research, and not all the cars shown in the video have the ground truth value. The total number of cars evaluated in the system and the actual speed value is 38.

3.2 Detection: YOLOv5

The limitation of this research is using only some of the vehicles shown in the video; the only vehicle detected is the car. Pre-trained YOLOv5s, YOLOv5m, YOLOv5l, and

YOLOv5x, are used for detecting cars. The pre-trained models were trained in the COCO dataset. Among 80 classes of the COCO dataset, the class used only cars. This pre-trained model is utilized due to the small number of datasets. Our dataset contains 449 images for training, 56 for validating, and 57 for testing. The labeling dataset is illustrated in Fig. 3.



Fig. 3. Dataset labelling (left) detected car in the testing dataset (right).

In this research, the authors use the architecture of YOLOv5. YOLOv5 is divided into three main parts: Backbone, Neck, and Head. In the backbone, CSPDarknet53 is used. Four pre-trained models were trained in a custom dataset with 16 batches and 100 epochs. For the research experiment, we train four different models of YOLOv5: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. To evaluate their detection performance, we calculate the precision and recall value. Then, to evaluate the time assuming or computational complexity, we estimate the average frame rate of the processing video. The precision and recall result are written in Fig. 4.

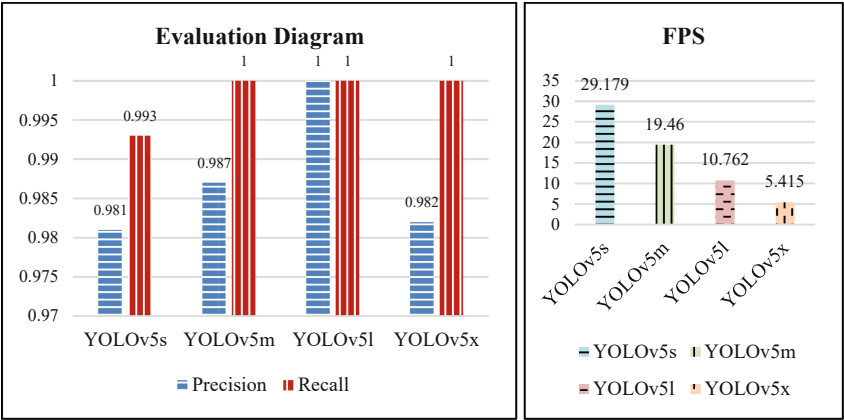


Fig. 4. Evaluation diagram (left) FPS diagram (right).

From the experiment, the highest precision and recall is reached by YOLOv5l of about 100% in both precision and recall. YOLOv5s have the lowest computation complexity, with a frame rate value reaching 29.179 fps. However, YOLOv5s have the lowest

precision and recall, 98.1% and 99.3%, respectively. In this research, we tend to use the YOLOv5l model since have the best performance, 100% of precision and recall, and the frame rate, 10.762, is better than YOLOv5x with a fps of about 5.415.

3.3 Tracking: DeepSORT

DeepSORT or Simple Online and Realtime Tracking with a Deep Association Metric is used for tracking the vehicle after detected. In DeepSORT, there are at least three steps: tentative, confirmed, and deleted. The whole process is included in Fig. 5 The cars detected using YOLOv5 would be tracked using Kalman Filter to predict the tracks of the bounding box from the past frame to the next frame. The tracker will be deleted if the track is lost when less than five frames are reached. The scenario of the tracking is included in dimensional state space:

$$(u, v, \gamma, h, \dot{x}, \dot{y}, \dot{\gamma}, \dot{h}), \quad (1)$$

where (u, v) is the bounding box position, γ is the aspect ratio, h is height, and $(\dot{x}, \dot{y}, \dot{\gamma}, \dot{h})$ are the respective velocities in image coordinates. The detection is based on their two metrics using verified tracks and matching cascade. The first metric is Mahalanobis distance, and the second metric is feature metrics generated using a pre-trained CNN model for bounding box appearance descriptor; in this research, mobilenet architecture is applied. Unmatched tracks and detections will be passed to the next filter using the Hungarian algorithm with the IoU cost matrix.

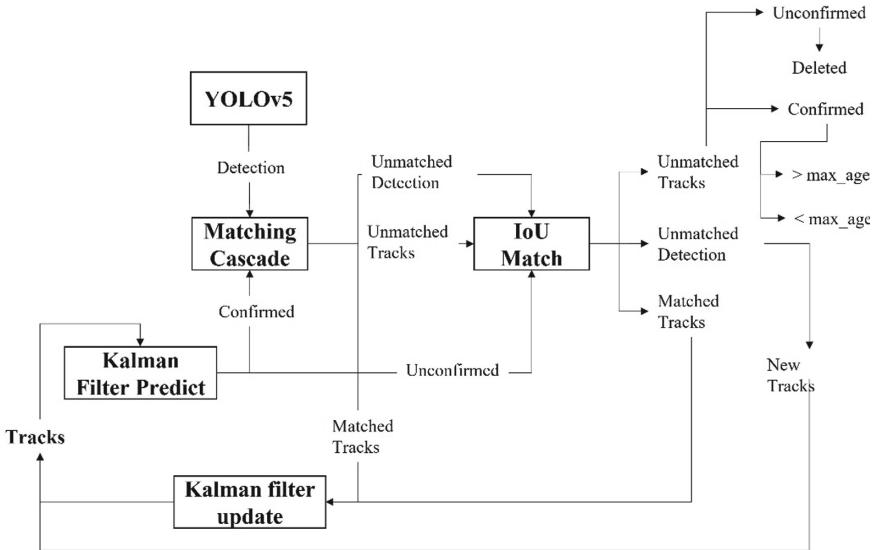


Fig. 5. DeepSORT workflow diagram

3.4 Speed Estimation

For estimating vehicle speed, pre-trained model YOLOv5s since they have the lowest time complexity. The following equation computes the vehicle speed measurements:

$$t = \frac{N}{fps}, \quad (2)$$

$$Speed = \frac{x \times 3.6}{t}, \quad (3)$$

where N is the number of frames, and fps is the frame rate. For obtaining the distance of the vehicle, four Region of Interest (RoI) is localized. In this experiment, two vehicles' directions are computed both toward and away from the camera. The distance between two regions x in each direction was measured by a meter laser, which is 12 m and 10 m. For the detail, look at Fig. 6. Furthermore, the elapsed time is calculated from the vehicles entering the first RoI and out of the second one for each direction based on generated centroid from the middle of the bounding box. The experiment result is also illustrated in Fig. 6. Including toward the camera, away from the camera, and both. Since the road segment has a roadblock in the center, the RoI can work adequately in each direction. Moreover, even if vehicles are in either direction, the system for both vehicles can operate.

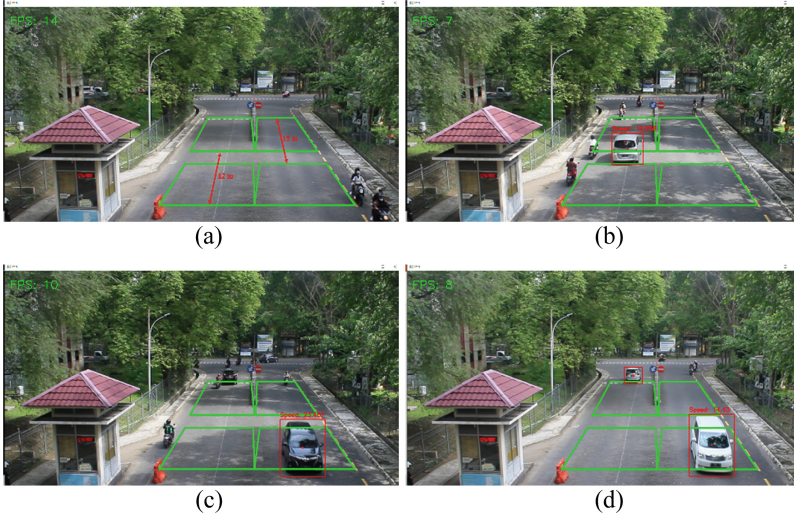


Fig. 6. Vehicle speed estimation: **a** Region of Interest (RoI) setting, **b** toward the camera, **c** away from the camera, and **d** both.

The number of ground truth data obtained from radar guns is 38 data. Table 1 contains vehicle speed results by the proposed system, speed ground truth by speed gun, and the difference between them. The car's speed is at most 34 km/h since the road is between the crossroads and the entrance gate of the college. The most considerable difference

between the system and ground truth is 11.593 km/h. The metrics utilized for evaluation are Root Mean Square Error (RMSE) and Mean Absolute Error (MAE).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{r}_i - r_i)^2}, \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{r}_i - r_i|, \quad (5)$$

where, $\hat{r}_i$ is the prediction rating and r_i is the actual rating in data set, n is the number of samples, and e is the model error composed of as e_i ($i = 1, 2, \dots, n$).

Table 1. Vehicle speed evaluation

Vehicle	Speed Estimation (km/h)			Vehicle	Speed Estimation (km/h)		
	System detection	Speed-gun detection	Difference		System detection	Speed-gun detection	Difference
1	32.995	29.000	− 3.995	20	35.564	34.000	− 1.564
2	24.379	26.000	1.621	21	22.412	25.000	2.588
3	24.23	19.000	− 5.23	22	17.31	24.000	6.69
4	22.251	20.000	− 2.251	23	20.788	18.000	− 2.788
5	22.883	21.000	− 1.883	24	23.175	18.000	− 5.175
6	16.335	21.000	4.665	25	18.006	23.000	4.994
7	24.209	20.000	− 4.209	26	20.005	23.000	2.995
8	21.595	22.000	0.405	27	24.691	23.000	− 1.691
9	18.335	17.000	− 1.335	28	23.829	25.000	1.171
10	23.968	20.000	− 3.968	29	21.59	27.000	5.41
11	21.595	22.000	0.405	30	29.024	32.000	2.976
12	23.968	20.000	− 3.968	31	21.415	25.000	3.585
13	17.241	17.000	− 0.241	32	20.088	23.000	2.912
14	30.706	29.000	− 1.706	33	24.182	28.000	3.818
15	31.351	30.000	− 1.351	34	25.738	21.000	− 4.738
16	24.948	24.000	− 0.948	35	30.588	31.000	0.412
17	30.531	24.000	− 6.531	36	19.407	21.000	1.593
18	28.279	24.000	− 4.279	37	20.503	19.000	− 1.503
19	32.714	26.000	− 6.714	38	19.407	31.000	11.593

The result reaches an RMSE of 3.926 and an MAE of 3.155. The result is excellent but still needs to be improved. The proposed study and the previous research are shown in

Table 2. It is shown that the proposed method’s result of this study has the minimum error. However, the dataset used differs from the previous study, so the RMSE comparison is not totally equal.

Table 2. Vehicle speed comparison result

References	Method	RMSE
Tang et al. [22]	YOLOv2, 3D SCT, Vehicle speed estimation	4.096
Kumar et al. [15]	Mask RCNN, SORT, Image rectification, Vehicle speed estimation	9.540
Kumar et al. [15]	Mask RCNN, DeepSORT, Image rectification, Vehicle speed estimation	10.100
Huang [13]	Faster RCNN, Histogram-based Tracking, Speed conversion	8.609
Proposed method	YOLOv5, DeepSORT, Vehicle speed estimation	3.926

4 Conclusion

We proposed the vehicle speed estimation method based on three steps: vehicle detection using YOLOv5, multi-object tracking using DeepSORT, and vehicle speed measurement. The authors build a primer traffic video dataset with ground truth value obtained from speed gun. Four YOLOv5 pre-trained models are trained using the custom dataset that our build contains 449 images for training, 56 for validating, and 57 for testing and comparing the precision, recall, and fps value. The experiment’s highest precision, 100%, and recall, 100%, are reached by YOLOv5l with 10.762 fps. For the vehicle speed estimation result, we reach RMSE 3.926 and MAE 3.155 through 38 data. The results are excellent. This research helps the law enforcer manage the speed limit in traffic. However, the dataset developed only contains slow vehicles speed, at most 34 km/h. In future research, we suggest including an automatic calibration technique to minimize error and append high vehicle speed.

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